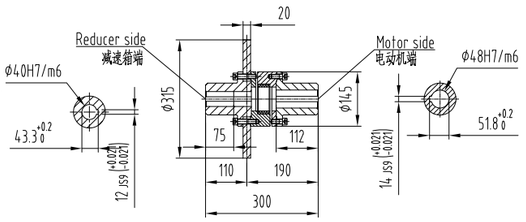
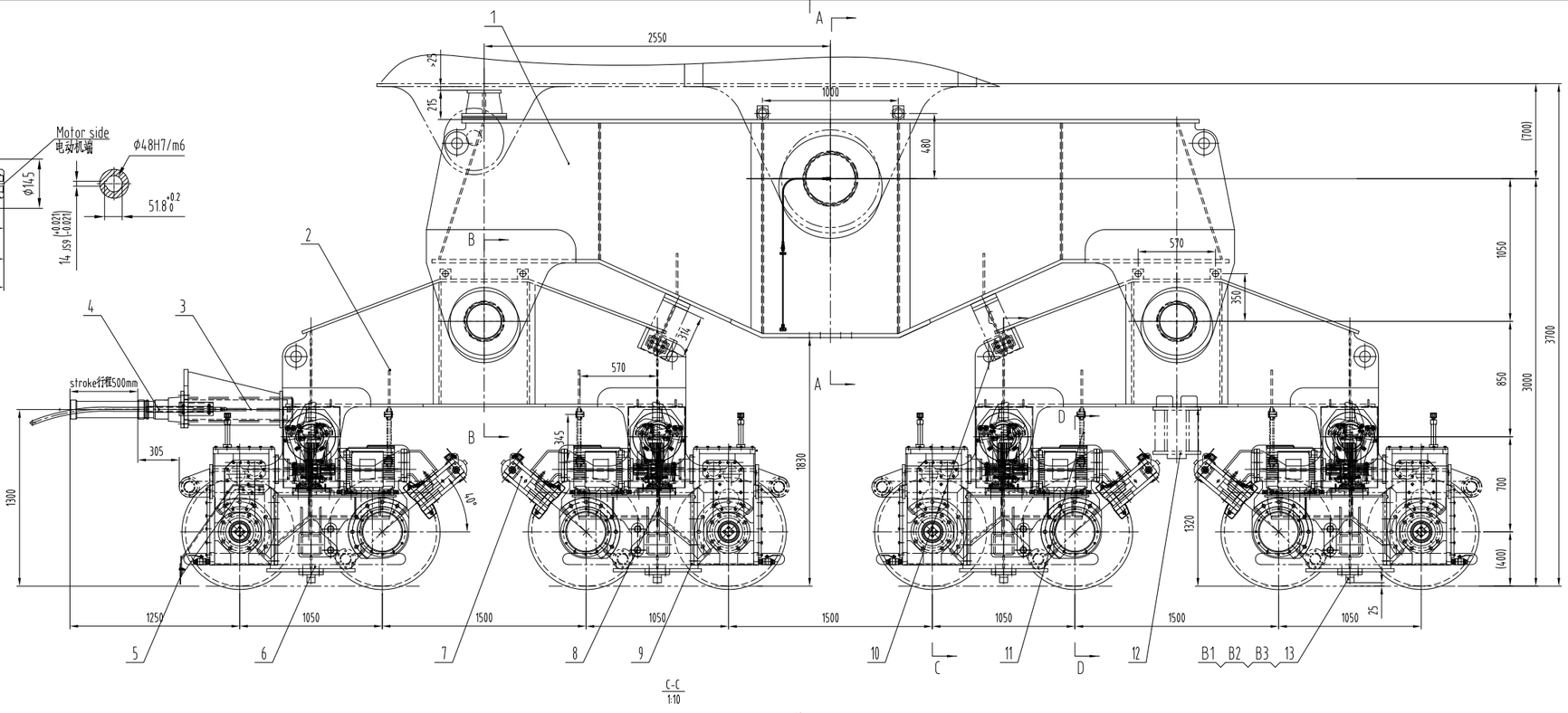
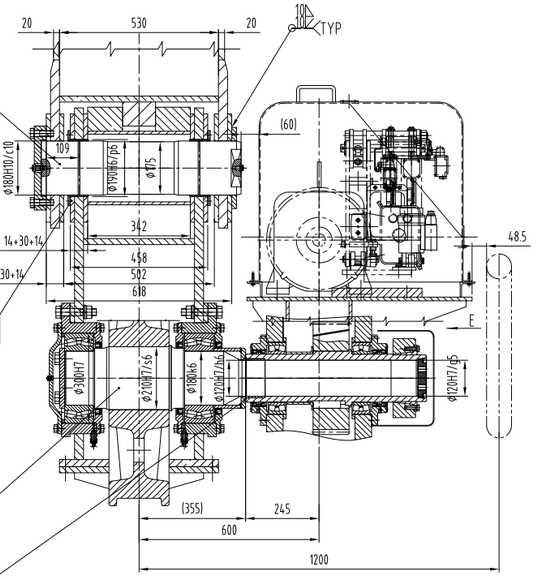
Coupling details
1:5

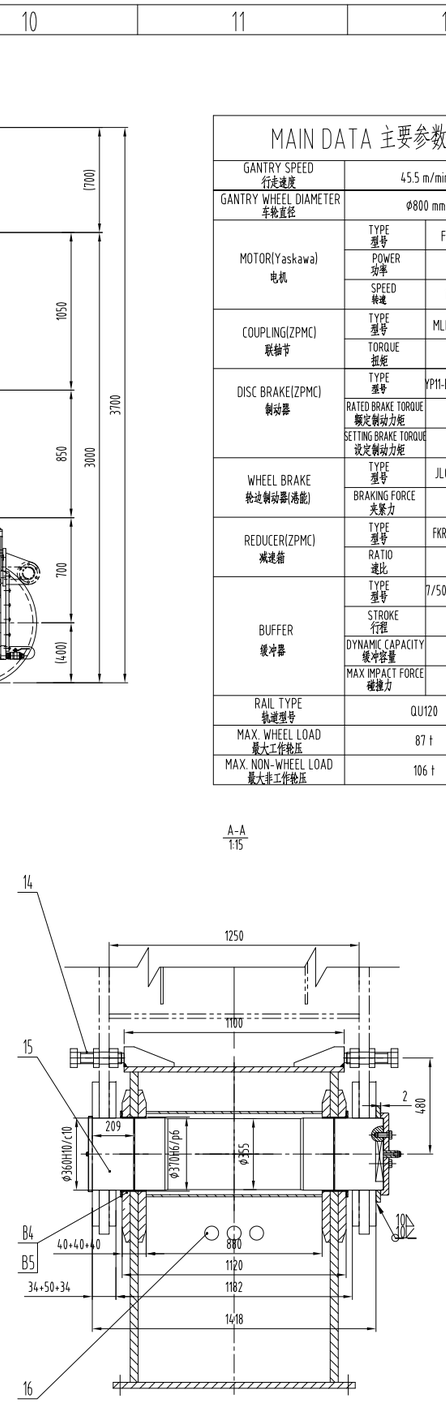




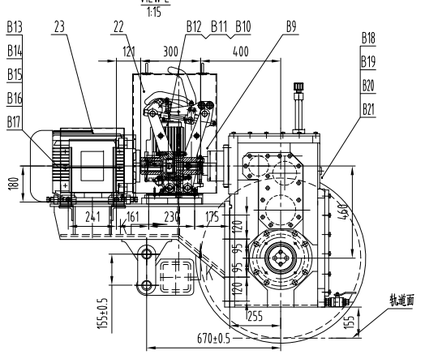
C-C
1:10



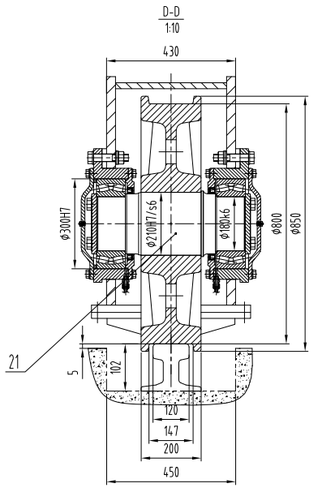
A-A
1:15



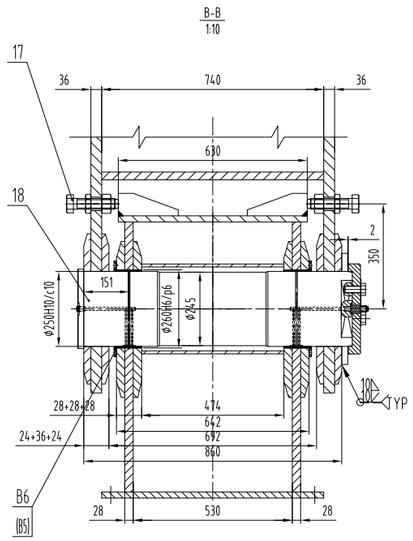
VIEW E
1:15



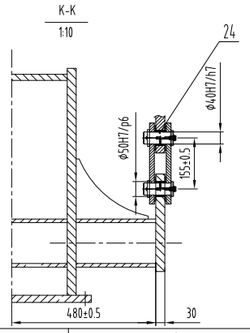
D-D
1:10



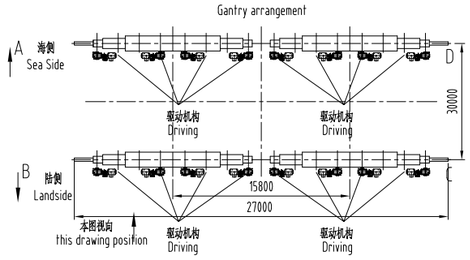
B-B
1:10



K-K
1:10



大车行走机构布置图
Gantry arrangement



Technical Requirements

1. Dispose the four travelling units of the crane according to the View of Travelling Mechanism Arrangement.
2. Tolerance requirement of wheel arrangement should be followed before shop assembly and general assembly:The centre planes of wheels rolling on a common rail diverging from the theoretic axis of wheels: less than ±1mm for the two wheels on one bogie, less than ±3mm for all the wheels of one corner, less than ±5mm for all the wheels on a common rail. The axes of the wheel bores must have an angular deviation less than ±0.08° from its theoretical axis, in the horizontal plane.
3. While fitting coupling, the angular displacement should be less than 0.1° , the radial misalignment should be less than 0.05mm, and the axial displacement should be less than 2mm.
4. All the bearing and bearing pin joints should be filled with appropriate grease during assembly. For driving wheels running in should be done.
5. The adjusting bolts are revolved outside before the whole installing in order to prevent from bumping badly,it should be tighted after the machine run on the quay.
6. GTF180AF should be installed from non-driving side to driving side.
7. The shrink disc should be set according to the reducer's manual.
8. Assured the surface of the railway shall be cleanness, without soldered dot and scraggy when the Gantry be dragging or hauling. In order to avoid damnification of the wheel tread.
9 .The enactment of motor brake should be set by Main data at the top right coner .

技术要求

1. 起重机共有四组行走机构，应按图示大车行走机构布置图布置。
2. 在车间及总装前车轮的排装要求：同一轨道上的车轮中心面同车轮理论中心线的偏差，每个台车上的2个车轮≤±1mm，每个角上的所有车轮≤±3mm，每根轨道上的全部车轮≤±5mm。轮孔中心线同其理论中心线之间在水平面内的角度偏差≤±0.08°。
3. 安装联轴器时角向偏移小于0.1°，径向偏移小于0.05mm，轴向偏移小于2mm。
4. 装配时，所有轴承均应加满润滑脂，主动车轮应跑合。
5. 调节装置的调节螺栓在过程中必须拿下来，防止被损坏，在用户码头待桥吊安装调试完毕后此调节螺栓需拧紧。
6. GTF180AF的安装方向，是由非驱动侧穿向驱动侧。
7. 减速箱上锁紧盘，必须严格按照其使用说明书进行安装与拆卸。
8. 在基地码头轨道上施行时应保证轨道面的清洁,无焊渣等凹凸不平现象,以免损伤大车车轮踏面。
9. 制动器的设定力矩应严格按照本图参数表内参数设定。

MAIN DATA 主要参数

GANTRY SPEED 行走速度		45.5 m/min
GANTRY WHEEL DIAMETER 车轮直径		φ800 mm
MOTOR(Yaskawa) 电机	TYPE 型号	FEK C21 OU
	POWER 功率	20 kW
	SPEED 转速	1750 rpm
COUPLING(ZPMC) 联轴节	TYPE 型号	MLPK6-315×20
	TORQUE 扭矩	630 Nm
DISC BRAKE(ZPMC) 制动器	TYPE 型号	YP11-ED220-50-315x20
	RATED BRAKE TORQUE 额定制动力矩	260 Nm
	SETTING BRAKE TORQUE 设定制动力矩	210 Nm
WHEEL BRAKE 轨道制动器(港能)	TYPE 型号	JLQ115.00-KH
	BRAKING FORCE 夹紧力	115 kN
REDUCER(ZPMC) 减速箱	TYPE 型号	FKR460.96.DIB-00
	RATIO 速比	96.734
BUFFER 缓冲器	TYPE 型号	7/500BQD140-517
	STROKE 行程	500 mm
	DYNAMIC CAPACITY 缓冲容量	280 KJ
	MAX IMPACT FORCE 碰撞力	700 KN
RAIL TYPE 轨道型号		QU120
MAX. WHEEL LOAD 最大工作轮压		87 t
MAX. NON-WHEEL LOAD 最大非工作轮压		106 t

REV	DATE	BY	DESCRIPTION
01	2012-5-30	Zhangsanjun	
02	2012-8-17	Zhangsanjun	

设计/DSGN	校核/CHKD	审核/EXAM	图幅/FORMAT
日期/DATE	日期/DATE	日期/DATE	A1
材料/MAT 部件总图Ass.	比例/SCALE 1:20	图名/DRAWING NAME 大车行走机构 gantry drive mechanism	
单重/WEI 135000			
ZPMC 上海振华港机公司	图号/DRAWING NO. J526A0600	版本/REV 0	页号/PAGE 1

010102

明 细 表
SUBSIDIARY LIST

*The quantity and weight of this list is to one set
*此表为一套的数量和重量

4 set/crane, 31000 kg/set, 124000 kg/crane

设计：

校对：

PROJECT NAME or CODE: Slilanka CICT 12 STS Cranes			SUB-PART NAME: 大车行走机构		SUB DRAW NO: J526A0600	
序号 NO.	图号或标准 SUB-NO. or STDD	名称与规格 NAME & SIZE	材料或型号 MTRL or STYLE	每套数量 QTY/SET	单重 PIECE WEIGHT	总重 TOTAL WEIGHT
1	J526A0601	Main Equalizer 大平衡梁	Welded 焊接件	1	8231	8231
2	J526A0602	Small Equalizer 小平衡梁	Welded 焊接件	2	1934	3868
3	J526A0612	Buffer Device 缓冲器装配	Assembly 组件	1	281	281
4	J526A0613	Buffer Safety Device缓冲器安全装置	Assembly 组件	1	8.7	8.7
5	GTE03	Cleaner 清扫器	Assembly 组件	1	24	24
6	J526A0603	Driving Bogie 主动台车	Welded 焊接件	4	788	3152
7	J526A060700	Wheel brake ass. 夹轮器装配	Assembly 组件	4	184	736
8	J526A0611	Motor base 电机支座	Assembly 组件	4	182	728
9	J526A0614	Enclosure护栏	Welded 焊接件	4	81	324
10	GTW1a-314/570	Steady Device 稳定装置1	Assembly 组件	2	68	
11	GTW2a-345/570	Steady Device 稳定装置2	Assembly 组件	2	28	
12	J526A0632	Jacking-Up support顶升支座	Welded 焊接件	2/12台机		
13	GTC02	Anti-break block 断轴保护块	Welded 焊接件	4	24.4	97.6
14	J526A062100	Adjustable device 1 调节装置1	Assembly 组件	1	33	33
15	GTF360A	Shaft 轴装配360*1418/209-D	Welded 焊接件	1	1186	1186
16	J526A063300	Electric collocation 电气布管图	Assembly 组件	1		
17	J526A062200	Adjustable device 2 调节装置2	Assembly 组件	2	13.6	27.2
18	GTF250A	Shaft 轴装配250*860/151-D	Welded 焊接件	2	358.5	717
19	GTF180AF	Shaft 轴装配180*618/109-D	Assembly 组件	4	149	596
20	J526A060400	Driving wheel Ass.主动车轮装配	Assembly 组件	4	856.3	3425.2
21	J526A060500	Driven Wheel Ass.从动车轮装配	Assembly 组件	4	797.4	3189.6
22	J526A0618	Motor Cover 电动机罩壳	Assembly 组件	4	13	52
23	GTL05b	Brake cover 制动器罩壳	Assembly 组件	4	17	68
24	J526A062000	Pin Assembly 销轴装配	Assembly 组件	4	11.4	45.6

明 细 表
SUBSIDIARY LIST

*The quantity and weight of this list is to one set
*此表为一套的数量和重量

4 set/crane, 31000 kg/set, 124000 kg/crane

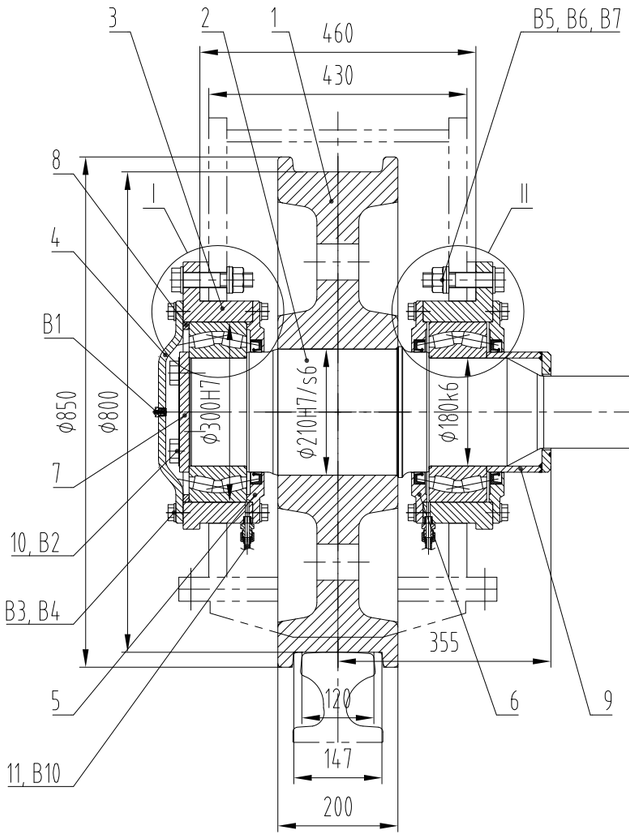
设计：

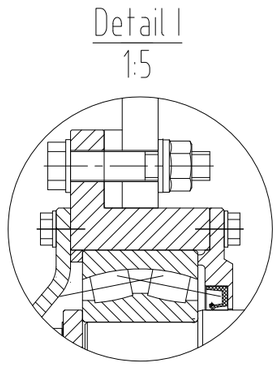
校对：

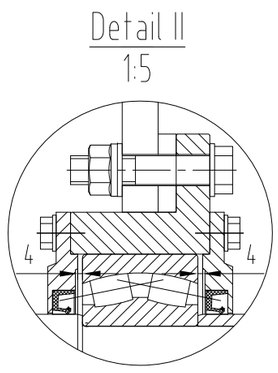
PROJECT NAME or CODE: Slilanka CICT 12 STS Cranes			SUB-PART NAME: 大车行走机构		SUB DRAW NO: J526A0600	
序号 NO.	图号或标准 SUB-NO. or STDD	名称与规格 NAME & SIZE	材料或型号 MTRL or STYLE	每套数量 QTY/SET	单重 PIECE WEIGHT	总重 TOTAL WEIGHT
B1	GB5782-86	Bolt 螺栓 M16*70	8.8S Dacro达克罗	32	0.14	4.48
B2	ZN6175-T-SL	Nut 螺母 M16	8S Dacro达克罗	32	0.03	0.96
B3	GB97.1	Washer 垫圈 16	200HV Dacro达克罗	32	0.01	0.32
B4	GS 360*120*12	Sleeve 轴套	CSB-08G	2	5.7	11.4
B5	GB/T 70.3	Bolt 螺钉M8*25	Ste.Stl.不锈钢	104		0
B6	GS 250*85*8	Sleeve 轴套	CSB-08G	4	3.4	13.6
B7	GS 180*60*6	Sleeve 轴套	CSB-08G	8	2.4	19.2
B8	GB3452.1-82	Seal O型橡胶密封圈132x3.55G	Rubber 橡胶	4		0.1
B9	MLPK6-315x20	Coupling 联轴节	ZPMC	4	16.7	66.8
B10	YP11-315*20-ED220-50	Brake 制动器	ZPMC	4	95	380
B11	GB/T 5783	Bolt 螺栓M12*45	Ste.Stl. 不锈钢	16	0.03	0.48
B12	GB/T 855	Washer 垫圈12	Ste.Stl. 不锈钢	16		
B13	FEK C21 OU	Motor 电动机	Yaskawa	4	185	740
B14	GB5782-86	Bolt 螺栓 M12*60	Ste.Stl.不锈钢	16	0.14	2.24
B15	ZN6177.1-SL	Nut 螺母 M12	Ste.Stl.不锈钢	16	0.03	0.48
B16	GB97.1	Washer 垫圈 12	Ste.Stl.不锈钢	16	0.01	0.16
B17	AS2-14	Washer 调整垫圈14	OCr18Ni9-304	16组		
B18	FKR460.96.D1B-00	Reducer 减速箱	ZPMC	4	570	2280
B19	GB/T 5782	Bolt 螺栓M20*85	8.8S Dacro 达克罗	32	0.25	8
B20	ZN6175-T-SL	Nut 螺母M20	8S Dacro 达克罗	32	0.02	0.64
B21	GB/T 97.1	Washer 垫圈20	200HV Dacro达克罗	64		

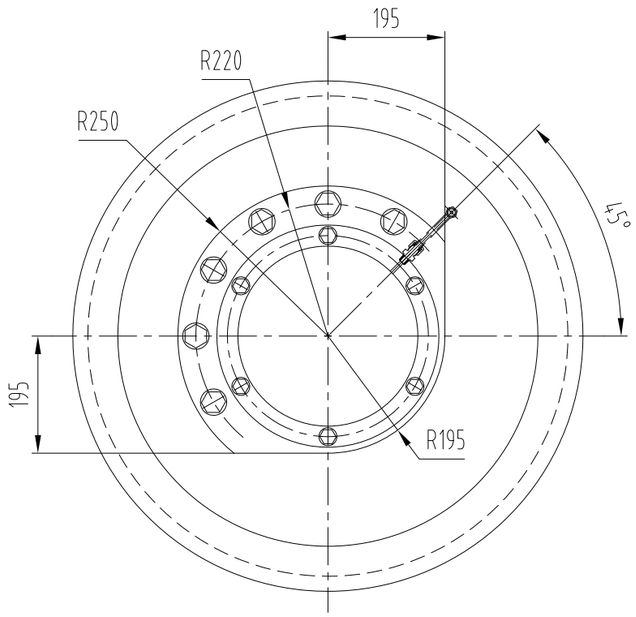
规格 Size	等级 Grade	预紧力(kN) Pretightening Force	预紧力矩(Nm) Pretightening Torque	预紧系数 Coefficient
M24	8.8 S	148.4	710	K=0.2

REV	DATE	BY	DESCRIPTION
S01	2012-5-30	Zhangsanjun	
S02	2012-8-17	Zhangsanjun	









技术要求

1.装配时所有轴承内均应加满润滑脂，装配完毕走轮必须跑合。
2.同一轨道上的车轮中心面同车轮理论中心线的偏差，每个台车上的2个车轮≤±1mm，每个角上的所有车轮≤±3mm，每根轨道上的全部车轮≤±5mm。轮孔中心线同其理论中心线之间在水平面内的角度偏差≤±0.08°。
3.螺栓的预紧力和预紧力矩见附表。
4.B3安装时需在螺纹表面涂乐泰胶防松。

Technical Requirement

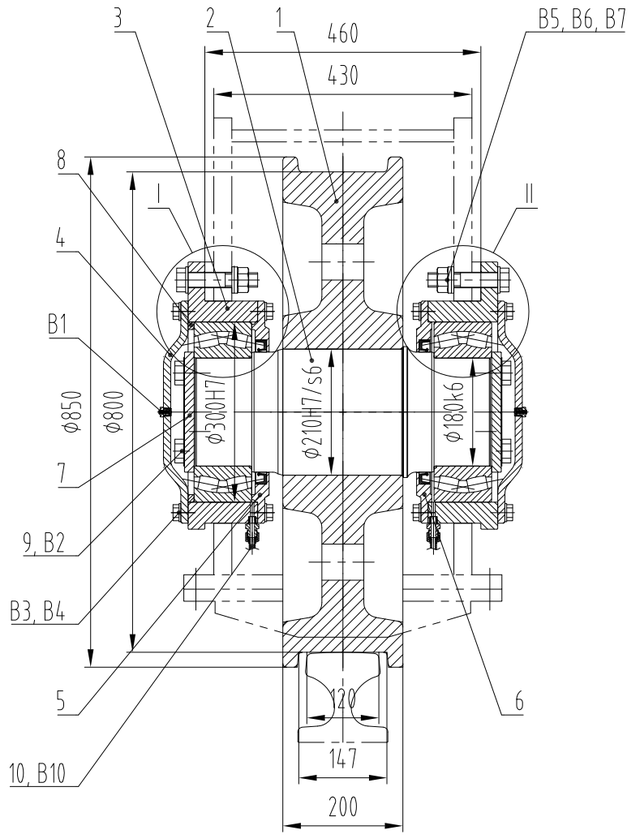
1. During assembly all bearing should be filled with appropriate grease, wheel running should be done.
2. The centre planes of wheels rolling on a common rail diverging from thetheoretic axis of wheels: less than ±1mm for the two wheels on one bogie, less than ±3mm for all the wheels of one corner, less than ±5mm for all the wheels on a common rail. The axes of the wheel bores must have an angular deviation less than ±0.08° from its theoretical axis, in the horizontal plane.
3. Pretightening force and torque of bolts refer to the table.
4. Item B3 should be srayed Loctie on the thread when fixing.

NO.	SUB-NO. OR STDD	VER	NAME & SIZE	MTRL	QTY	EACH(Kg)	TOTAL(Kg)	NOTE
B10	JISB0203		平头油嘴 PT1/4" Nipple	外购成品 Pur.	2	0.0	0.0	'黄铜'
B9	GB/T288-94		轴承 23136 Bearing	外购成品 Pur.	2	27.1	54.2	'180×300×96 'Cr>=1030
B8	HG4-692-67		油封 PD200×240×18 Oil seal	外购成品 Pur.	3	0.0	0.0	
B7	GB97.1-85		平垫圈 24 Washer	200HV	24	0.0	0.0	'Dacro'
B6	ZN6175-T-SL		六角凸缘自锁防松螺母 M24 Nut	8H	12	0.0	0.0	'Dacro'
B5	GB5782-86		螺栓 M24×120 Bolt	8.8	12	0.6	7.2	'Dacro'
B4	GB97.1-85		平垫圈 16 Washer	200HV	24	0.0	0.0	'Dacro'
B3	GB5783-86		螺栓 M16×40 Bolt	8.8	24	0.1	2.4	'Dacro'
B2	GB5783-86		螺栓 M24×50 Bolt	8.8	3	0.3	0.9	'Dacro'
B1	FQZR0.25-5		放气嘴 R1/4" Nipple	外购成品 Pur.	1	0.0	0.0	
11	J526A060410	0	润滑管接头 Lubrication connector	组件 Ass.	2	2.0	4.0	
10	J526A060409	0	防松垫圈 Washer	Q235-A	1	0.1	0.1	
9	J526A060408	0	轴套 Sleeve	Q235-A	1	5.9	5.9	
8	J526A060407	0	卡环 Ring	Q235-A	1	0.9	0.9	
7	J526A060406	0	轴端挡板 End Plate	Q235-A	1	3.7	3.7	
6	J526A060405	0	透盖2 Cover 2	Q235-A	1	8.0	8.0	
5	J526A060404	0	透盖1 Cover 1	Q235-A	1	8.2	8.2	
4	J526A060403	0	端盖 Cover	Q235-A	1	10.5	10.5	
3	J526A060402	0	轴承座 Bearing Pedestal	ZG270-500	2	36.9	73.8	
2	J526A060401	0	主动车轮轴 Driving Shaft	35CrMo	1	185.5	185.5	
1	GWF2-800x120-J526A	0	走轮φ800x120 Wheel	42CrMo	1	491.0	491.0	

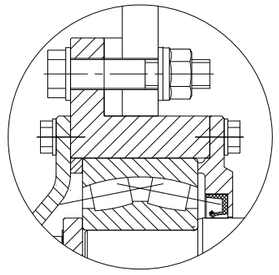
描述及日期/Description & Date	设计/DSGN	校对/CHKD	审核/EXAM	图幅/FORMAT
	日期/DATE	日期/DATE	日期/DATE	A3
	材料/MAT 组件 Ass.	比例/SCALE 1:10	图名/DRAWING NAME 主动车轮装配 Driving Wheel Assembly	
	单重/WEI (Kg) 856.3			
修改标志/Sign	ZPMC 上海振华重工集团	图号/DRAWING NO. J526A060400	版本/REV 0	页号/PAGE 1

规格 Size	等级 Grade	预紧力(kN) Pretightening Force	预紧力矩(Nm) Pretightening Torque	预紧系数 Coefficient
M24	8.8 S	148.4	710	K=0.2

REV	DATE	BY	DESCRIPTION
S01	2012-5-2	Zhangsanjun	
S02	2012-8-17	Zhangsanjun	

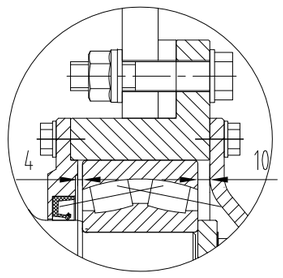


Detail I
1:5

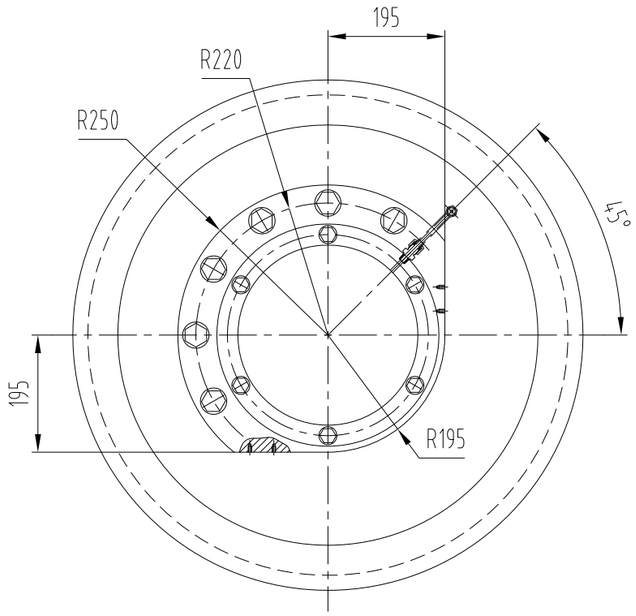


出油方向
Grease out

Detail II
1:5



出油方向
Grease out



技术要求

1.装配时所有轴承内均应加满润滑脂，装配完毕走轮必须跑合。
2.同一轨道上的车轮中心面同车轮理论中心线的偏差，每个台车上的2个车轮≤±1mm，每个角上的所有车轮≤±3mm，每根轨道上的全部车轮≤±5mm。轮孔中心线同其理论中心线之间在水平面内的角度偏差≤±0.08°。
3.螺栓的预紧力和预紧力矩见附表。
4.B3安装时需在螺纹表面涂乐泰胶防松。

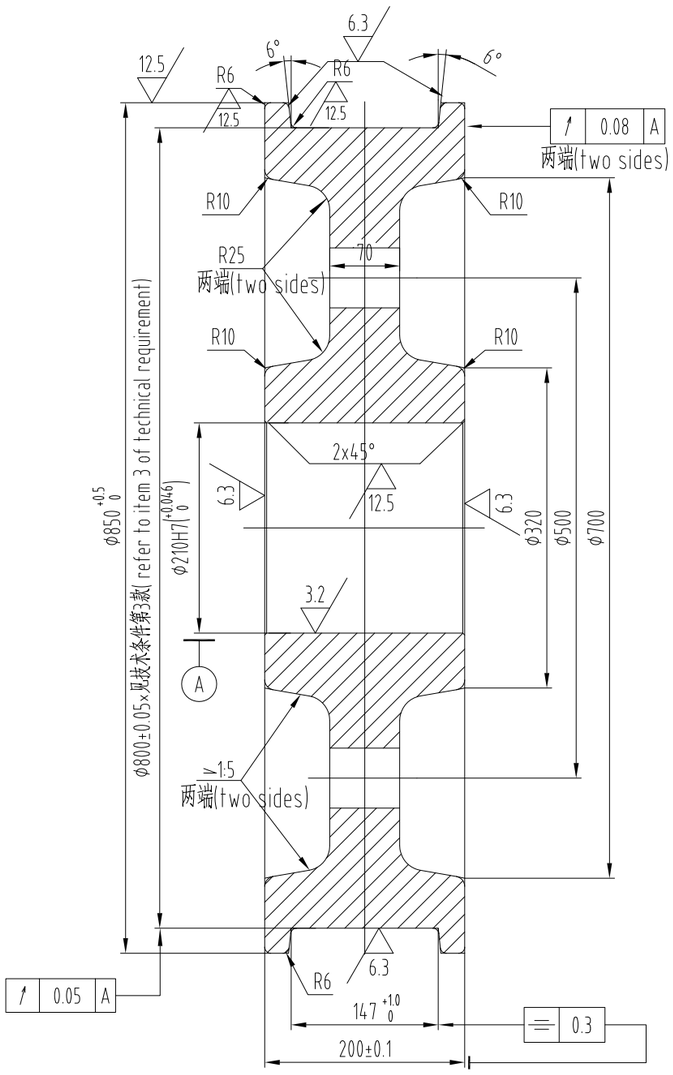
Technical Requirement

1. During assembly all bearing should be filled with appropriate grease, wheel running should be done.
2. The centre planes of wheels rolling on a common rail diverging from thetheoretic axis of wheels: less than ±1mm for the two wheels on one bogie, less than ±3mm for all the wheels of one corner, less than ±5mm for all the wheels on a common rail. The axes of the wheel bores must have an angular deviation less than ±0.08° from its theoretical axis, in the horizontal plane.
3. Pretightening force and torque of bolts refer to the table.
4. Item B3 should be srayed Loctie on the thread when fixing.

B10			平头油嘴 NPT1/4" Nipple	外购成品 Pur.	2	0.0	0.0	'黄铜'
B9	GB/T288-94		轴承 23136 Bearing	外购成品 Pur.	2	27.1	54.2	'180x300x96 'Cr>=1030Kn
B8	HG4-692-67		油封 PD200×240×18 Oil seal	外购成品 Pur.	2	0.0	0.0	
B7	GB97.1-85		平垫圈 24 Washer	200HV	24	0.0	0.0	'Dacro'
B6	ZN6175-T-SL		六角凸缘自锁防松螺母 M24 Nut	8H	12	0.0	0.0	'Dacro'
B5	GB5782-86		螺栓 M24×120 Bolt	8.8	12	0.6	7.2	'Dacro'
B4	GB97.1-85		平垫圈 16 Washer	200HV	24	0.0	0.0	'Dacro'
B3	GB5783-86		螺栓 M16×40 Bolt	8.8	24	0.1	2.4	'Dacro'
B2	GB5783-86		螺栓 M24×50 Bolt	8.8	6	0.3	1.8	'Dacro'
B1	FQZR0.25-5		放气嘴 R1/4" Nipple	外购成品 Pur.	2	0.0	0.0	
10	J526A060410	0	润滑管接头 Lubrication connector	组件 Ass.	2	2.0	4.0	
9	J526A060409	0	防松垫圈 Washer	Q235-A	2	0.1	0.2	
NO.	SUB-NO. OR STDD	VER	NAME & SIZE	MTRL	QTY	EACH(Kg)	TOTAL(Kg)	NOTE

8	J526A060407	0	卡环 Ring	Q235-A	1	0.7	0.7	
7	J526A060406	0	轴端挡板 End Plate	Q235-A	2	3.7	7.4	
6	J526A060405	0	透盖2 Cover 2	Q235-A	1	8.0	8.0	
5	J526A060404	0	透盖1 Cover 1	Q235-A	1	8.2	8.2	
4	J526A060403	0	端盖 Cover	Q235-A	2	10.5	21.0	
3	J526A060402	0	轴承座 Bearing Pedestal	ZG270-500	2	36.9	73.8	
2	J526A060501	0	从动车轮轴 Driven Shaft	35CrMo	1	117.5	117.5	
1	GWF2-800x120-J526A	0	走轮φ800x120 Wheel	42CrMo	1	491.0	491.0	
NO.	SUB-NO. OR STDD	VER	NAME & SIZE	MTRL	QTY	EACH(Kg)	TOTAL(Kg)	NOTE

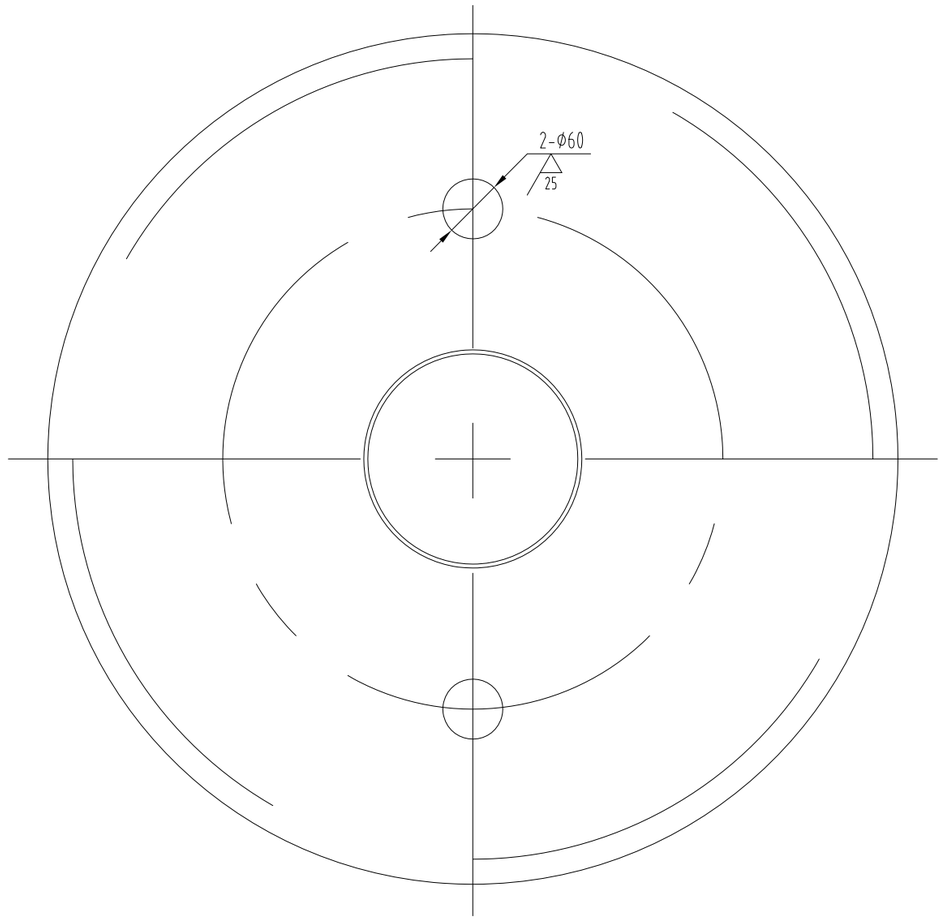
描述及日期/Description & Date	设计/DSGN	校对/CHKD	审核/EXAM	图幅/FORMAT
	日期/DATE	日期/DATE	日期/DATE	A3
	材料/MAT 组件 Ass.	比例/SCALE 1:10	图名/DRAWING NAME 从动车轮装配 Driven Wheel Assembly	
	单重/WEI (Kg) 797.4			
修改标志/Sign	ZPMC 上海振华重工集团	图号/DRAWING NO. J526A060500	版本/REV 0	页号/PAGE 1



技术要求

1. 制作时按公司内控标准(高于ASTM A504标准)制作: 踏面硬度HRC50~56, 硬化层有效深度≥15mm,15mm深处硬度>HRC35.
2. 车轮锻件按GB12361-90和GB12362-90技术条件验收.
3. 被动轮的车轮踏面尺寸公差为∅800±0.1,本图为主动车轮.
4. 加工时,应在侧面上作好主从动车轮记号.

Others(其余)



Technical Requirement

1. Surface heat treatment shall be according to the ZPMC standard which is more critical than the ASTM A504;the hardness of the wheel tread shall be HRC50~56,the effective hardness depth shall not be less than 15mm, hardness at 15mm deepth shall not be less than HRC35.
2. The wheel body should be checked and accepted according to the national forging standard GB12361-90 and GB12362-90.
3. The diameter tolerance of driven wheel is ∅800±0.1.This drawing indicates the driving wheel ,the diameter tolerance is ∅800±0.05.
4. Distinguish the driving or driven wheel by earmarking on the wheel side during manufacturing period.

描述及日期/Description & Date	设计/DSGN		校对/CHKD	图幅/FORMAT
	日期/DATE		日期/DATE	A3
	材料/MAT 42CrMo	比例/SCALE 1:6	图名/DRAWING NAME 走轮∅800 Wheel∅800	
	单重/WEI(Kg) 491			
修改标志/Sign	ZPMC 上海振华重工集团		图号/DRAWING NO. GWF2-800-J526A	版本/REV 0 / 页号/PAGE 1

010330d